

Agent's Utility CARA:

$$u(y) = -e^{-py}$$

Let: $y \sim N(m, v)$

Then

$$\begin{aligned} E[u(y)] &= E[-e^{-py}] \\ &= -E[e^{-py}] \end{aligned}$$

$$E[e^{-py}] = E[e^{qy}], \text{ where } q = -p$$

So

$$E[e^{qy}] = \int_{-\infty}^{\infty} e^{qy} \frac{1}{\sqrt{2\pi v}} e^{-\frac{(y-m)^2}{2v}} dy$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi v}} e^{(qy - \frac{(y-m)^2}{2v})} dy$$

$$qy - \frac{(y-m)^2}{2v} = -\left[\frac{y - (m + vq)}{2v}\right]^2 + \frac{mq + vq^2}{2}$$

So

$$= \int_{-\infty}^{\infty} e^{\frac{mq + vq^2}{2}} \frac{1}{\sqrt{2\pi v}} e^{-\frac{(y - (m + vq))^2}{2v}} dy$$

$$= e^{\frac{mq + vq^2}{2}} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi v}} e^{-\frac{(y - (m + vq))^2}{2v}} dy}_1$$

$$= e^{\frac{mq + vq^2}{2}}$$

$$\begin{aligned} E[e^{qy}] &= e^{\frac{mq + vq^2}{2}} \\ &= e^{-pm + vp^2/2} \end{aligned}$$

and $E[u(y)] = -e^{-pm + vp^2/2}$

Certain equivalent:

$$u(CE) = E[u(y)]$$

$$-e^{-pCE} = -e^{-p(m - vp/2)}$$

$$CE = m - vp/2$$

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Define θ_{ik} : latent developer - licensee productivity match

$$\theta_{ik} | F_{it} \sim N(\mu_{ik,t}, \frac{1}{\pi_{ik,t}})$$

$\mu_{ik,t}$ = belief mean about θ

$\pi_{ik,t}$ = belief precision about θ